Extending an Outer-Ear Simulator into a Middle-Ear Model

MUE 610 Psychoacoustics — AI Build Lab · JINGLAN HUANG

Middle-ear circuit after Pascal, Bourgeade, Lagier & Legros, *J. Acoust. Soc. Am.* **104**(3), 1509–1516 (1998), Figure 1 and Table I.

Interactive simulator https://hhhjinglan-a.github.io/ear-simulator/
Code and full documentation https://github.com/hhhjinglan-A/ear-simulator
The live page recomputes every number in this report in the browser and lets you move any parameter. This document is a fixed snapshot of it.

Headline result. The model reproduces the paper's own published curves to **1.34 dB rms in magnitude and 7.9° rms in phase** for the transfer function (Fig. 3), and **0.91 dB / 2.5°** for the input impedance (Fig. 2) — both inside the error of reading the scanned figures. The JavaScript port of the model agrees with the MATLAB original to **8.1×10^{-16}** relative error over 12 scenarios × 96 frequencies.

1 The circuit and how its topology was verified

The five subsystems form a **ladder network**, not a chain. "Series" and "shunt" describe where an element sits, not whether it dissipates: a series element carries all the volume velocity and the pressure it drops is subtracted from what continues on; a shunt element offers a parallel route back to the return, diverting velocity that then never reaches the cochlea. Only the resistances dissipate.

My own check against Figure 1. I checked the main series and parallel connections directly against Figure 1 myself: the cavity branches; that the eardrum-loss branch taps *before* C_te; the series C_is–R_is shunt branch; and R_co in parallel with the series R_h–L_h branch. I also confirmed that p_c is taken across the cochlear load, after L_v. That check caught an error in the schematic printed in an earlier draft of this report, which drew Z_eardrum at node 2, omitted Z_stapes altogether, and terminated the cochlear load at node 2 instead of node 3. The schematic below is the corrected one. The computation code did not contain this error — it had the ladder right — so no number, figure or conclusion anywhere in this report changes as a result.

```
                      node 1              node 2              node 3
p_t o---[Z_cavity]------o---[Z_ossicles]----o---[Z_stapes]-----o
     u_t                |                   |                  |
                  [Z_eardrum]           [Z_joint]         [Z_cochlea]  <- p_c
                        |                   |                  |
                       gnd                 gnd                gnd
```

Block	Elements (Fig. 1)	Position	Internal topology
Z_cavity	L_a 12 mH, C_cp 3.6 μF, R_a 20 Ω, R_cm 420 Ω, C_cm 0.35 μF	**series**	(L_a+C_cp+R_a) ‖ R_cm ‖ C_cm
Z_eardrum	R_ti1 200, C_ti1 0.5μF, C_ti2 0.3μF, R_ti2 105, L_ti 15 mH, R_ti3 12500, C_ti3 0.2μF	**shunt**, taps *before* C_te	R_ti1 + C_ti1 + ((C_ti2+R_ti2)‖L_ti) + (R_ti3‖C_ti3)

Z_ossicles	C_te 1.4 μF, L_te 40 mH, R_te 65 Ω	**series**	all in series
Z_joint	C_is 0.03 μF, R_is 170 Ω	**shunt**, taps *before* L_s	R_is + C_is
Z_stapes	L_s 8 mH, C_st, R_la, C_la, L_v 21 mH	**series**	all in series
Z_cochlea	R_co 1211 Ω, R_h 850 Ω, L_h 150 mH	**terminal load**	R_co ∥ (R_h + L_h)

The transformer Figure 1 does not draw

The paper states (p. 1510) that the eardrum-to-cochlea impedance match "is not represented in Fig. 1 but must be considered in the computation". Eq. (1) gives `T_r = lever × area ratio = 1 × 17 = 17`. All element values are already referred to the tympanic side, so the pressure across the referred cochlear load is multiplied by T_r **once**.

Four independent checks of the transcription

Check	Model	Independent derivation	Agreement
Compliance ↔ volume	C_cp+C_cm = 3.95 μF	$V\_c/(\rho_a c^2) = 5.5/(1.15e\text{-}3\cdot(3.5e4)^2) = 3.904e\text{-}6$	`1.2 %`
Inertance ↔ mass	L_s = 8 mH	$(m/A\_f^2)/T\_r^2 = (2.5e\text{-}3/0.032^2)/289 = 8.448e\text{-}3$	`5.3 %`
Cochlear resistance	R_co = 1211 Ω	0.35e11 Pa·s/m^3 (Zwislocki 1975) ÷ T_r^2 ÷ 10^5 = 1211 The 10^5 converts SI Pa·s/m^3 to the CGS acoustic units the table is printed in (1 Ω ≡ 1 dyn·s/cm^5).	`exact`
IEC 318 canal circuit inherited from the outer-ear assignment	5634.95 Hz, 27.0018 dB	**ngspice** simulation of the same circuit: 5636.8 Hz, 27.00 dB	`3.3e−4`

Checks 2 and 3 only work if the elements are *divided* by T_r^2, which is what licenses multiplying the output by T_r once. The cavity compliances are *not* referred — they sum directly to $V\_c/(\rho c^2)$ — correctly, since the cavity sits on the eardrum side.

2 Responses: outer ear, middle ear, combined

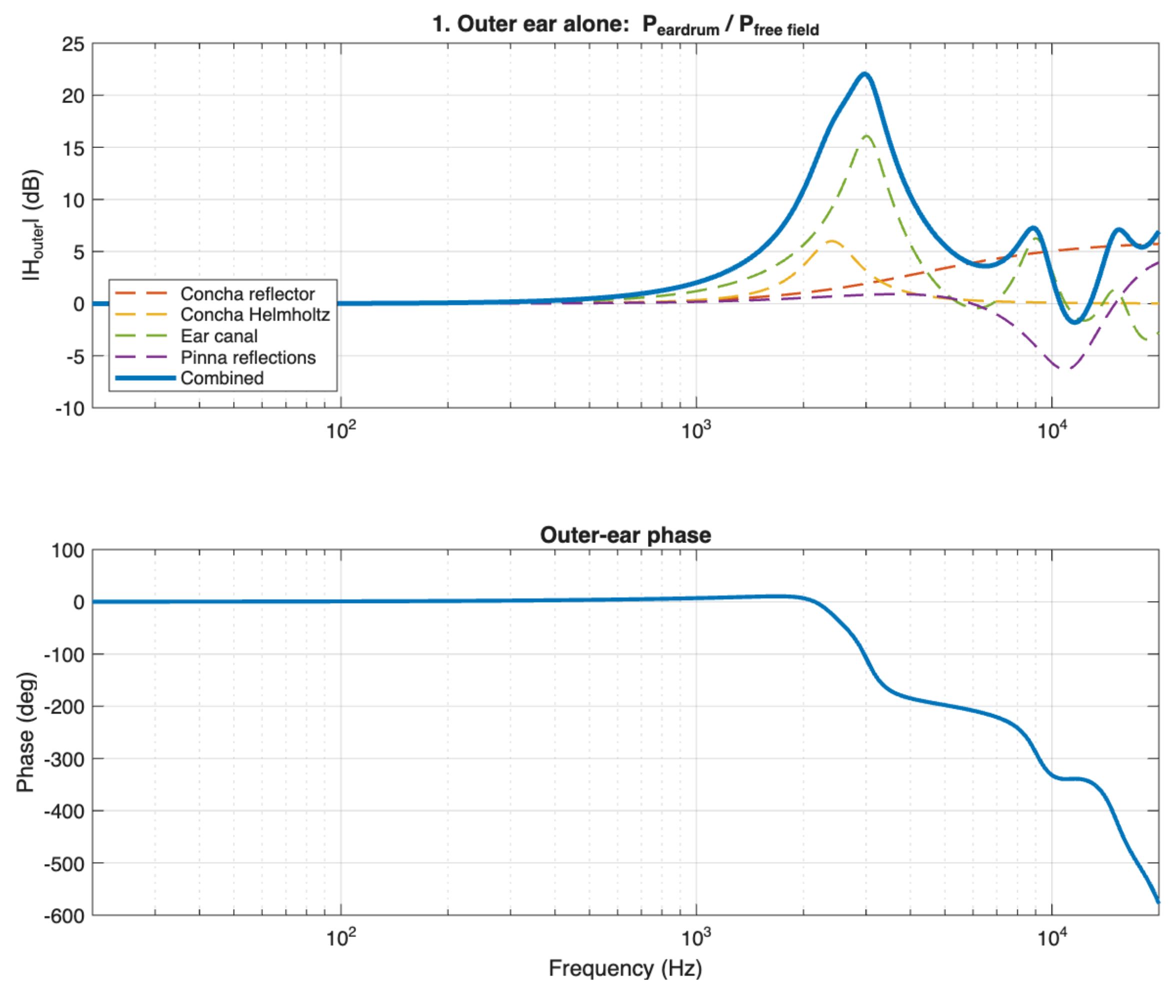

Figure 1. Outer ear alone: the four mechanisms and their product. Peak +22.0 dB at 2973 Hz.

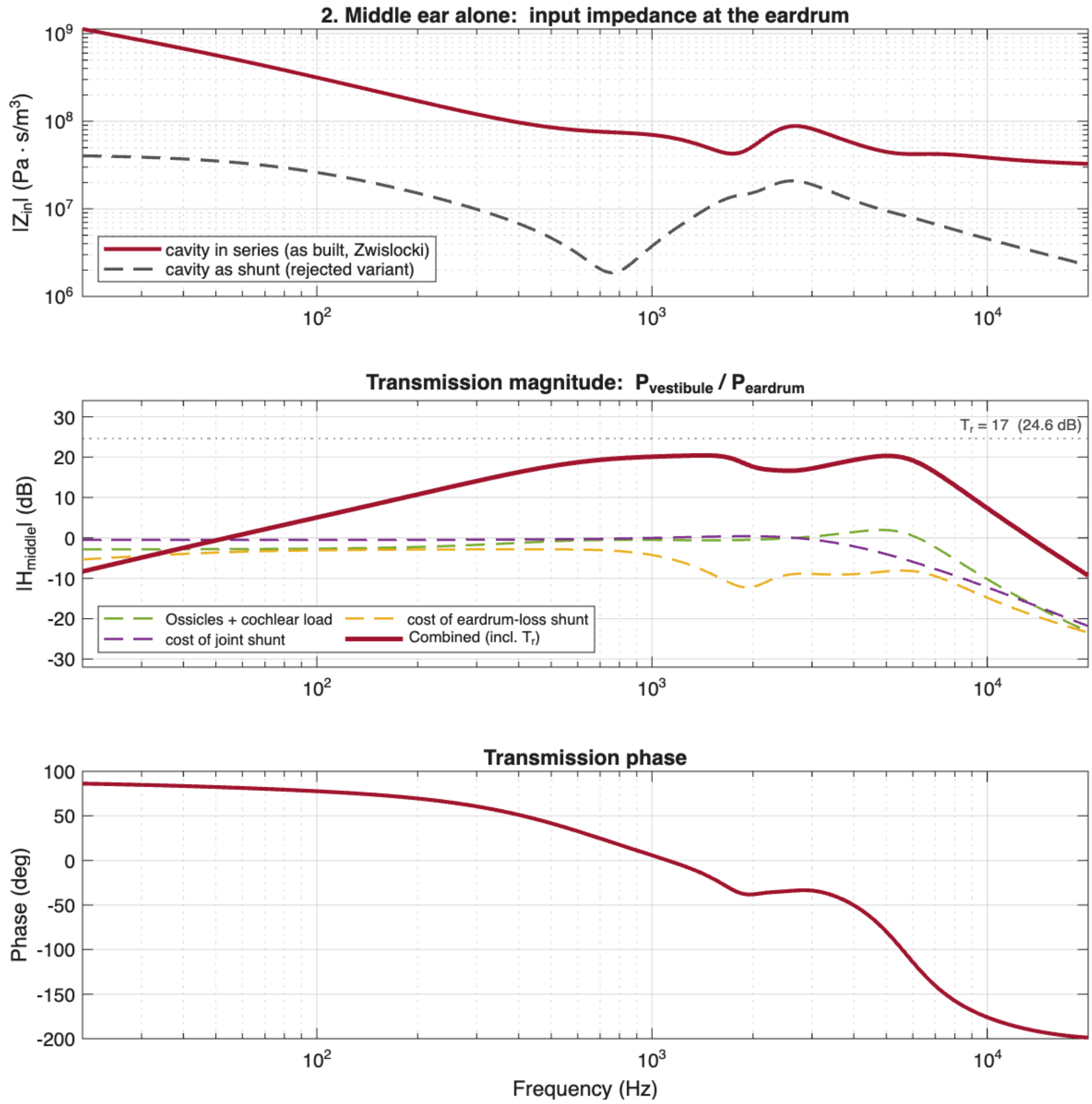

Figure 2. Middle ear alone. Top: input impedance $|z_t|$, falling from 1.13e9 Pa·s/m^3 at 20 Hz to its minimum 3.27e7 at 20 kHz, with a local minimum of 4.26e7 at 1738 Hz. Middle: transmission magnitude, peak +20.4 dB at 1423 Hz. Bottom: transmission phase.

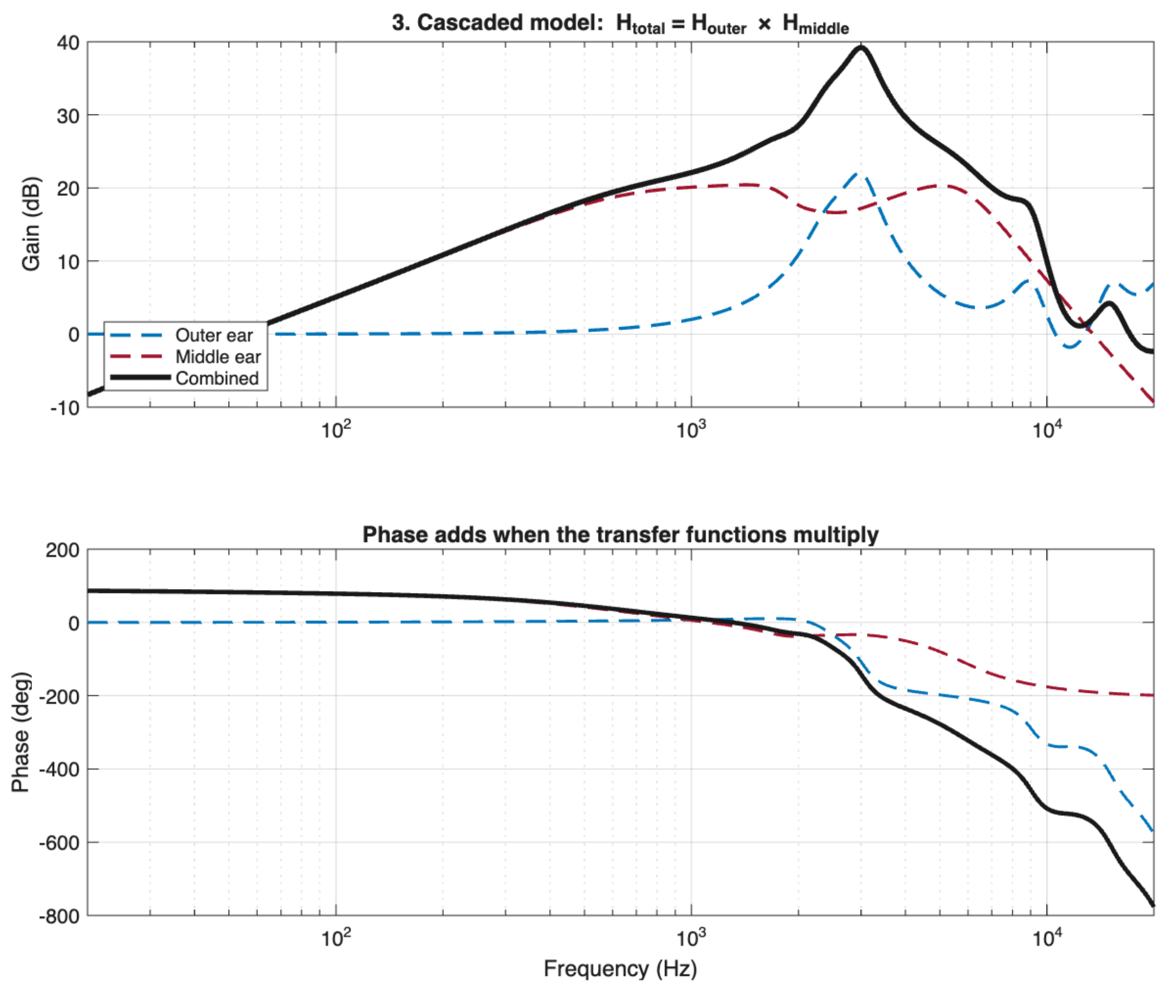

Figure 3. Cascade $H_{total} = H_{outer} \times H_{middle}$. Combined peak +39.2 dB at 2999 Hz.

Cascade assumption and loading

The product is only valid if the middle ear draws negligible volume velocity from the canal, i.e. $|z_t| \gg Z_{canal}$. Against the canal characteristic impedance ρc/A = 9.35e6 Pa·s/m^3 the ratio is **median 8.9, minimum 3.49** — not the “much greater than” the product assumes. It first drops below 5 at 1544 Hz and dips to **4.56 at 1738 Hz**, near the middle-ear resonance, recovers to about 6 at 4 kHz, then falls steadily to its **overall minimum 3.49 at 20 kHz**. The worst loading is therefore at the top of the band, not at the resonance. Nothing in the model corrects for this; the live page plots the ratio against frequency.

3 Three parameter experiments

All three start from the published defaults; one parameter changes at a time. MATLAB, the web app and this table use the same 4096-point log grid and linear interpolation and agree to the last digit.

ΔH_{middle} (dB)	100 Hz	500 Hz	1 kHz	2 kHz	4 kHz	10 kHz
E1 L_te +20 % — inertance	+0.010	+0.096	−0.036	−0.116	−0.137	**−1.957**
E2 C_is ×2 — compliance	−0.148	−0.101	+0.211	**+0.883**	+0.043	**−6.698**
E3 R_te ×2 — loss	−0.033	−0.414	**−0.471**	−0.321	−0.446	−0.060

	peak gain (dB)	peak frequency (Hz)	$\lvert z_t \rvert$ minimum (Hz)
baseline	20.43	1423	1738
E1	20.32	1410	1741
E2	21.03	1493	1723
E3	20.05	1443	1738

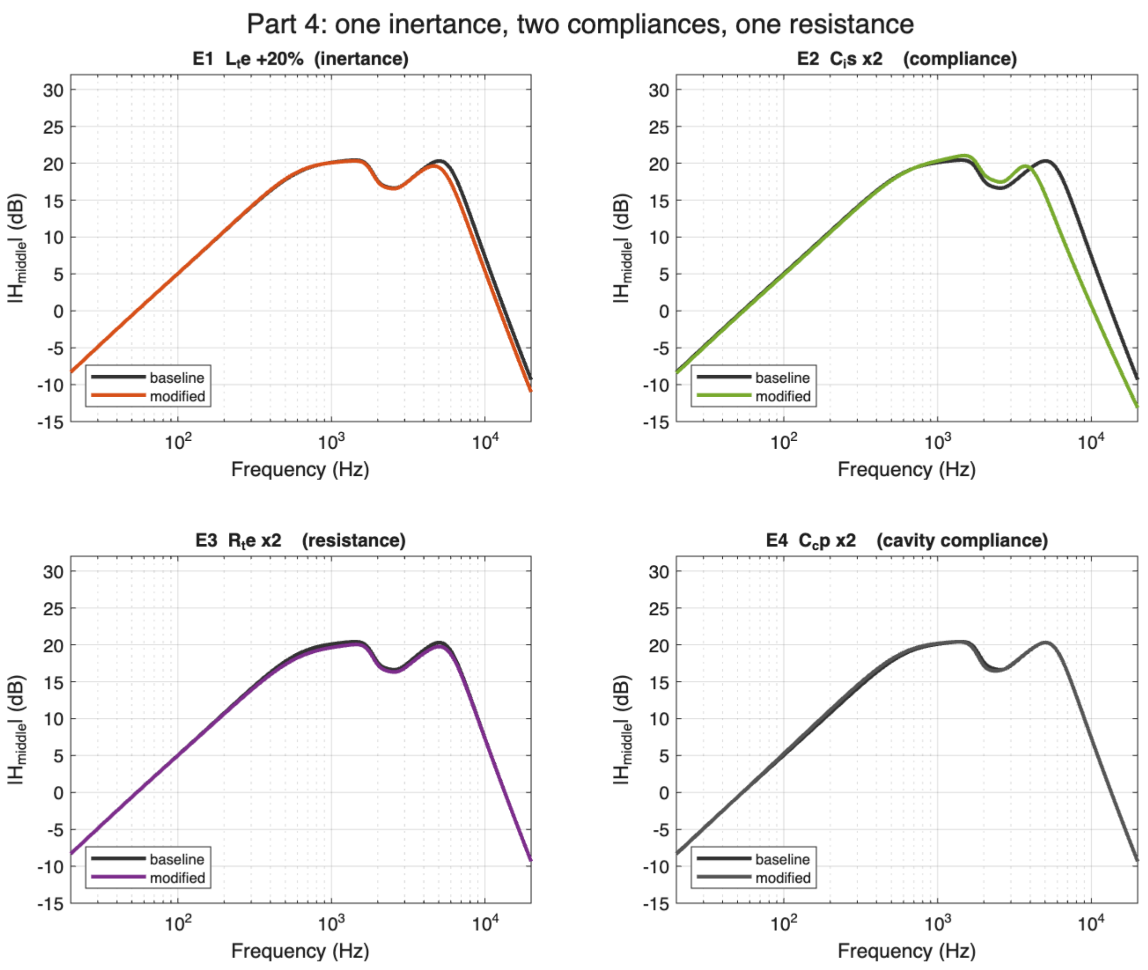

Figure 4. The three experiments plus a control (C_cp ×2).

My own predictions, written before running, and how they scored

Experiment	My prediction	Result	Verdict
E1 resonance	goes up	isolated L_te–C_te pair fell 8.7 %	wrong direction
E1 100 Hz	(answered with a correct statement about compliance, not the question asked)	+0.010 dB, unchanged	partly
E1 10 kHz	level goes up	−1.957 dB	wrong direction
E2 low frequency	yes, it will change	−0.148 dB, 15× E1's change	defensible
E2 high frequency	will rise	−6.698 dB at 10 kHz — but *+0.883 dB at 2 kHz*	wrong at the top, right about the middle
E3 which band	the mid band	−0.471 dB at 1 kHz vs ≈−0.05 dB at both ends	**correct**

> **What the errors had in common.** Both E1 mistakes and the E2 high-frequency mistake share one missing step: deciding *which impedance dominates in that band* before deciding which way the gain moves. At 100 Hz the compliance impedance is about 45× the mass impedance, so changing the mass does nothing; at 10 kHz the mass dominates instead. Once that step was supplied I applied it correctly and unaided in E3. The first answers are kept verbatim in the online report rather than replaced, because the progression is the point.

What the results corrected in the model's own explanation

- **E1.** The isolated L_te–C_te resonance (673 → 614 Hz, −8.7 %) is *not* the system peak (1423 → 1410 Hz, −0.9 %). Treating them as one number would be wrong by a factor of ten. Not predicted: 500 Hz *gained*, because the resonance lifted the response on its way down.
- **E2.** "A softer shunt always loses signal" is false. C_is also resonates with the stapes and vestibular mass; doubling it moved that resonance 5396 → 3815 Hz, which *raised* the 1–4 kHz region and the peak gain. This is the experiment that most directly tests whether the shunt topology is right.
- **E3.** The peak moved *up* 20 Hz. An earlier claim in this project that damping "drags the peak downward" came from a superseded circuit and was withdrawn.

4 Parameter provenance

Parameter	Classification	Basis
C_cp + C_cm	**Physically derived**	Eq. (2) $C = V/(\rho_a c^2)$ from the measured cavity volume; reproduces the printed values to 1.2 %.
L_s	**Physically derived**	Stapes mass 2.5 mg on a 3.2 mm² footplate, m/A_f² referred through T_r². Agrees to 5.3 %.
T_r = 17	**Physically derived**	Eq. (1): lever 1 × area 55/3.2 = 17.2, rounded to 17 in Table I.
R_co = 1211 Ω	**Measured**	Cochlear acoustic impedance 0.35e11 Pa·s/m³ (Zwislocki 1975), referred through T_r² and converted to CGS acoustic units ($\div 10^5$). Not derivable from anatomy.
R_ti1–3, C_ti1–3, L_ti	**Fitted**	The paper states these were "estimated using a minimization algorithm (Nelder and Mead, 1965)". There is no anatomical object corresponding to "cell 2".
R_h, L_h	**Fitted**	Same Simplex minimisation; the cochlear model form was "defined before testing many solutions".
Child cavity volume 2.8 cm³	**Assumed — no source**	No measured paediatric middle-ear *cleft* volume was found. Published child figures of 0.4–1.3 cm³ are *ear-canal* equivalent volumes, a different cavity. Largest single uncertainty in the extension.
All otitis-media magnitudes	**Assumed**	Directions are literature-supported; magnitudes are order-of-magnitude guesses.

An open question, not a correction to the paper. Eq. (7) prints the annular-ligament capacitance coefficients in farads, giving C_la = 0.5067 F at rest. In this analogue 1 F ≡ 1 cm^5/dyn, so that is ~10^5× the entire middle-ear cavity — effectively a short circuit. Read as *microfarads* it becomes a real stiffness, and only that reading reproduces the paper's own figures (1.34 vs 3.56 dB rms). The microfarad reading is used, but this is indirect evidence about the authors' intent and should be confirmed against Lutman & Martin (1979).

5 Limitations

1. **The cascade is approximate, and worst at the top of the band.** $|z_t|/Z_{canal}$ is 4.56 at 1738 Hz, near the middle-ear resonance, where the middle ear absorbs the most power and genuinely loads the canal; but it reaches its true minimum **3.49 at 20 kHz**. Either way the cascade never has the headroom the product assumes.
2. **Lumped elements cannot describe the ear above about 3 kHz.** The circuit assumes one pressure and one volume velocity per node; the real tympanic membrane breaks into modes. The paper itself notes (p. 1511) that above 2.5 kHz the incudomalleal joint introduces a phase angle it treats as stiff.
3. **The two models do not share a boundary condition.** The outer-ear model terminates its canal in a *rigid* eardrum, not in the z_t computed here, and nothing feeds back from stapes to concha.

4. **The outer-ear stage already over-predicts and the cascade inherits it** — about +22 dB near 3 kHz where a real ear measures +15 to +20 dB, a 2–7 dB offset carried into the combined figure.
5. **Strictly linear.** The acoustic reflex and level-dependent annular ligament (Eqs. 3, 6, 7) are documented but not implemented.

6 Graduate extension: otitis media in a small child

> **Exploratory simulation, not a validated clinical model.** Every magnitude is assumed. The child baseline is kept separate from the effusion because they rest on different evidence.

Circuit values are **derived** where they can be: the remaining air volume gives C_cp and C_cm through Eq. (2), and a fluid layer of depth *t* on the drum gives an added inertance ρt/A_t. **Assumed**: the child cleft volume, the fill fraction, the fluid depth, and three multipliers on damping, drum stiffening and aditus swelling. Nothing was tuned to reproduce a target hearing loss.

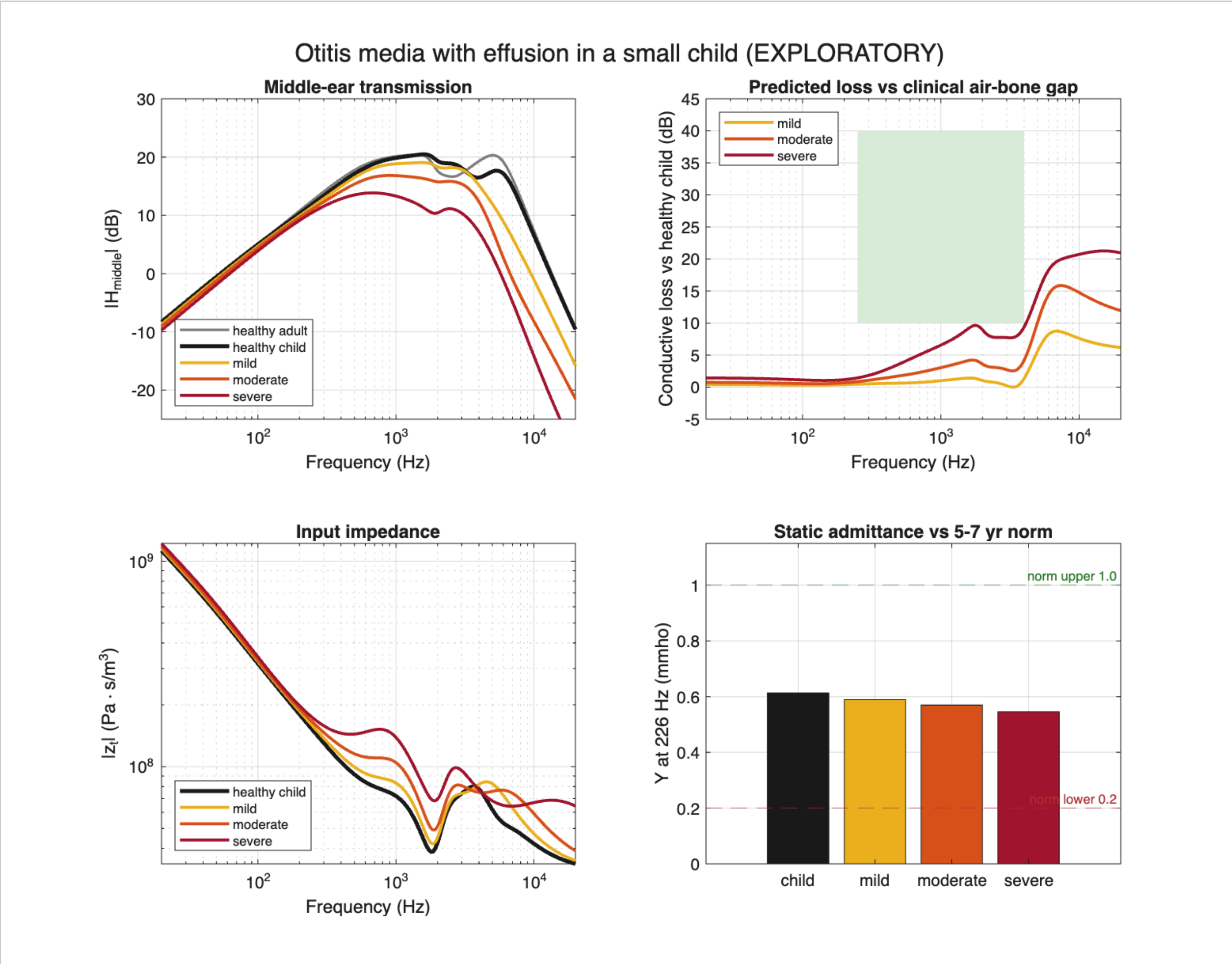

Figure 5. Healthy adult, healthy child and three effusion severities.

Ear	4PTA loss vs healthy child	flatness	Y(226 Hz)	resonance
healthy child (V_c 2.8 cm³)	–	–	0.61 mmho	1817 Hz
mild (φ 0.30, t 0.2 mm)	1.0 dB	1.4 dB	0.59	1850 Hz

moderate (ϕ 0.60, t 0.5 mm)	3.1 dB	3.4 dB	0.57	1881 Hz
severe (ϕ 0.90, t 1.0 mm)	7.1 dB	8.2 dB	0.55	467 Hz

This extension does not reproduce clinical otitis media, and that is reported rather than tuned away. The direction of the loss and its flat configuration are right, but the magnitude is far too small (1–7 dB against a clinical air-bone gap of 10–40 dB), the static admittance never reaches type B (<0.2 mmho), and the resonance frequency moves the *wrong way* for mild and moderate.

The cause was measured, not guessed. R_cm = 420 Ω sits *in parallel* across the whole cavity block, capping its impedance no matter how much air the fluid displaces: healthy child 417.5 Ω; severe effusion with ten times less air 419.9 Ω; the same case with R_cm removed 40 038 Ω. The circuit is **structurally insensitive to cavity volume**. Filling the cleft is also a *stiffening*, which raises a resonance, whereas real OME lowers it because the fluid mass-loads the drum. Fixing this needs a different element, not a different number, which is why the assumed severities were not adjusted to close the gap.

7 AI use, and where I corrected it

One assistant was used throughout: Claude (Opus 5), as coding copilot. The most consequential error changed the modelled pressure gain at 4 kHz by more than 10 dB.

Step	Who	What happened
1	AI	With no paper available, built the five subsystems as a simple series/parallel chain, and invented a transformer ratio of about 22 from an assumed 1.3 lever ratio.
2	**Me**	**Rejected it.** Told the assistant the circuit is a *ladder* network, and supplied Eq. (1) giving $T_r = 17$.
3	AI	Rebuilt as a ladder, but still wrong in three places that a prose description cannot fix.
4	AI	*Self-check:* flagged that it had written "Values are Figure 1 / Table I of Pascal et al." into source it had never read, and downgraded it to "provenance unverified".
5	**Me**	**Supplied the paper.** Nothing below could have been found without it.
6	AI	Reading Fig. 1, found three errors: the eardrum-loss branch is one series chain with two parallel sub-blocks, not three parallel cells; it taps *before* C_{te}; and p_c is measured across the cochlear load *alone*. Agreement with the paper improved from ~9.7 to 1.34 dB rms; 4 kHz moved from 8.1 dB to 19.3 dB against the paper's 20.8 dB.
7	AI	*Self-check:* noticed its own web-vs-MATLAB comparison reported a perfect "0.00e+0"; every comparison against NaN is false, so a NaN would have scored perfectly. Hardened it to count non-finite values.
8	**Me**	**Asked whether the outer-ear assignment had actually been integrated.** It had not — its engine had been copied in without its test suite. Its 20 checks now run first.
9	**Me**	**Rejected two explanations:** that "series elements are losses and shunt elements are dead ends", and that magnitude is "insensitive to topology". Checking its own records, the assistant found the wrong circuit had been off by 12.7 dB at 4 kHz — magnitude had caught it immediately — and withdrew the claim.
10	**Me**	**Rejected post-hoc labelling.** The written "predictions" for E1–E3 had been composed after the results were known but labelled "before running". They are now labelled as reasoning written afterwards, and one genuinely blind prediction was committed to git before its run as an audit trail.
11	**Me**	**Checked the circuit against Figure 1 myself and found the report's schematic wrong.** It drew $Z_{eardrum}$ at node 2, omitted Z_{stapes}, and terminated the cochlear load at node 2. I specified the correction: $Z_{eardrum}$ at node 1, Z_{joint} at node 2, Z_{stapes} between node 2 and the load, load and p_c at node 3. The assistant had drawn this figure correctly in three other places and coded it correctly in both languages, and had not noticed that the one figure in the submitted report disagreed with all of them.

What this shows. The assistant was reliable at arithmetic, at running its own tests, and at flagging claims it could not support. It was unreliable whenever it had to guess at structure it had not seen: it produced a confident, plausible circuit that was wrong by more than 10 dB, and no amount of self-checking found that — only the source document did. Every substantive physics correction traces back to my supplying the source or rejecting an explanation.

8 Verification summary

Check	What it establishes	Result
`test_vsPaper`	the model matches the paper's Figs. 2 and 3	`1.34 / 0.91 dB · 7.9° / 2.5°`
`test_middleEar`	circuit algebra, units, transformer, response shape	`80 checks, 0 failed`
`test_audioIR`	the audio path equals the analytic transfer function	`23 checks, 0 failed`
`test_outerEar`	inherited outer-ear engine, incl. the ngspice cross-check	`all passed`
`tools/compare.js`	the web port equals MATLAB, in the complex plane	`8.1e−16 relative`

Magnitude and phase establish different things and are reported separately. Magnitude tests the element values and the transformer referral; phase adds independent evidence about topology and ordering, since a model could in principle be tuned to fit a magnitude curve with compensating values and still disagree in phase. Neither is blind to topology: when this circuit was wrong, its magnitude was off by 12.7 dB at 4 kHz.

9 Outstanding

1. The otitis-media extension does not reach clinical magnitude; the cause is located at R_cm.
2. The child cavity volume has no source.
3. Eq. (7)'s units remain an open question pending Lutman & Martin (1979).
4. That second paper is a scan with no text layer and has not been read, so its Table 1 versus Figure 12 discrepancy (R_D2 = 12 or 220 Ω; C_C = 0.65 or 0.6 μF) is unresolved. Those elements do not appear in this model, so no value has been silently chosen.
5. The non-linear block (Eqs. 3, 6, 7) is documented but not implemented; setting `p2.nonlinear = true` raises an explicit error rather than returning a linear result.